JSP

Java Server Pages.

- **JSP is a technology is used to create web application just like Servlet technology.**

- **It can be thought of as an extension to Servlet(Advanced version of Servlet) because it provides more functionality than servlet such as JSP tags, implicit objects , etc.**

- **A JSP page consists of HTML tags and JSP tags.**

- **The JSP pages are easier to maintain than Servlet because we can separate designing and development.**

Advantages of JSP over Servlet

- 1) **Extension to Servlet** : JSP technology is the extension to Servlet technology. We can use all the features of the Servlet in JSP. In addition to, we can use implicit objects, predefined tags , Custom tags etc in JSP, that makes JSP development easy.
- 2) **Easy to maintain** : JSP can be easily managed because we can easily separate our business logic with presentation logic. In Servlet technology, we mix our business logic with the presentation logic.
- 3) **Fast Development** : If JSP page is modified, we don't need to recompile and redeploy the project. The Servlet code needs to be updated and recompiled if we have to change the look and feel of the application.
- 4) **Less code than Servlet** : In JSP, we can use many tags such as jsp tags that reduces the code. Moreover, we can use implicit objects, etc.

JSP SCRIPTING ELEMENTS (JSP TAGS)

- **The scripting elements provides the ability to insert java code inside the jsp.**

- **There are three types of scripting elements:**

1. **Scriptlet Tag**
2. **Expression Tag**
3. **Declaration Tag**

EXPRESSION TAG:

- The code placed within JSP expression tag is written to the output stream of the response. So you need not write out.print() to write data. It is mainly used to print the values of variable or method.

- Syntax is as follows:

<%= statement %>

Example :

```
<html>
<body>
<%= "welcome to jsp" %>
</body>
</html>
```

- In this example, we are displaying a welcome message.
- Note: Do not end your statement with semicolon in case of expression tag.

DECLARATION TAG:

- The JSP declaration tag is used to declare fields and methods.

- Syntax is as follows:

<%! field or method declaration %>

Example :

- <html>
- <body>
- <%! int data=50; %>
- <%= "Value of the variable is:"+data %>
- </body>
- </html>

- In this example, we are displaying a welcome message.

EXAMPLES OF JSP TAGS

```jsp
<%@ page language="java" contentType="text/html; charset=UTF-8"
    pageEncoding="UTF-8"%>
<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

<!-- DECLARATION TAG : declaring variables and methods-->

<%!
    String name = "sheela";
    int age = 25;

    int a = 100;
    int b = 200;
%>

<!-- EXPRESSION TAG : output statements -->
<h1>
<%= "NAME : "+name+" AGE : "+age %>
<br><br>
<%= "Sum of "+a+"and "+b+" is : "+(a+b) %>
</h1>

</body>
</html>
```

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<!DOCTYPE html>
<html>
<head>
<meta charset="UTF-8">
<title>Insert title here</title>
</head>
<body>

<%!
    int cube(int n) {
        return n*n*n;
    }
%>
<h1>
<%= "Cube of 5 is : "+cube(5) %>
</h1>

</body>
</html>
```